



Docking markers for space

The European Space Agency is developing docking markers for satellites to allow them to interface each other in orbit. <https://digit.in/may19-71>



Robonaut to return to space

NASA's robotic test platform is going to return to the ISS, this time with the addition of legs that can grab rails. <https://digit.in/may19-72>

DOWN TO EARTH

The technologies developed for space exploration often end up translating into products that are useful to humans down on Earth as well. These are the practical benefits of the space age.

Aditya Madanapalle |
aditya@digit.in

INFORMATION TECHNOLOGY

The most widely used technology to emerge from NASA is the complementary metal oxide semiconductor (CMOS) sensor. This is the kind of

sensor that is used in smartphones and action cameras. These sensors are more compact and less power intensive than their predecessors, the charge coupled device CCD sensors, which were preferred in the early days of digital photography, which was also a result of NASA developments in the 1960s. In its early days, those who

advocated the advantages of CMOS technology had a hard time, as they were seen as competition to those who were experts in the field of CCD technology. The earliest companies to adopt the technology were Kodak and GoPro. Almost all digital cameras in the world now use CMOS sensors. The two biggest sensor manufacturers in the world, Sony and Samsung both license the technology from Caltech, which holds the patents and manages NASA's Jet Propulsion Laboratory (JPL). The cameras that you use in your phones were originally developed for smaller and less energy intensive instruments for spacecraft.

NASA engineers trying to make fuel flow efficiently in low gravity conditions, came up with the solution of using tiny particles of iron oxide to magnetise the fluid in 1960. More than half a century later, in 2012, Sony started using ferrofluids as a damper in their speakers, which is a kind of a brace that prevents the diaphragm from tearing. This allowed Sony to create slimmer speakers that they were able to accommodate in their televisions, home theatre systems, as well as movie theatres and recording studios. These ferrofluid based speakers improved the audio accuracy over traditional dampers, produced higher fidelity sound, and allowed for playback in higher volumes without distortion.

MATERIALS

Gold has various properties that make it ideal for astronomical instruments. It is really inert and does not deteriorate over time, and it is good for reflecting infrared wavelengths, making it suitable for use in telescopes that study really distant objects. However, gold is notoriously soft, and requires the addition of other metals such as nickel or cobalt to give it strength, but the gold loses its reflectivity in the process. NASA contractor Epner came up with a proprietary gold coating method, that drives current from a gold cyanide anode to a negatively charged cathode, which is to be plated. The individual gold atoms are